CS 700 Midterm Notes Mon 10/26, open book and notes, no electronics

1. Metrics (Lecture 2)

Good metrics are linear, reliable, repeatable, easy to use, consistent and independent. Reliable, repeatable, consistent and independent are required. Linear and easy to use are nice but not necessary.

MIPS $= \dfrac{n}{T_e \times 10^6}$, MFLOPS $= \dfrac{f}{T_e \times 10^6}$. MIPS fails reliable and consistent (CISC vs RISC, CPI). MFLOPS gives an integer program 0 and a FLOP isn't defined the same everywhere. SPEC normalizes to a basis machine and takes a geometric mean, which is the wrong mean for times.

Means metrics (clock rate, MIPS, MFLOPS) count what was done. Ends metrics (execution time) count what got accomplished. Execution time is the fundamental metric. Report wall clock and CPU time.

Speedup of system 2 over 1 is $S_{2,1} = \dfrac{T_1}{T_2} = \dfrac{R_2}{R_1}$. Relative change is $\Delta_{2,1} = \dfrac{T_1 - T_2}{T_2} = S_{2,1} - 1$. Precision $= \dfrac{\text{relevant retrieved}}{\text{retrieved}}$, recall $= \dfrac{\text{relevant retrieved}}{\text{all relevant}}$.

2. Summarizing data (Lecture 4)

Center

Mean $\bar{x} = \frac{1}{n}\sum x_i$, median is the middle sorted value (average the two middle ones when n is even), mode is the most frequent value or bin. Use the median when the data is skewed or has outliers, the mode for categorical data, and the mean when the total matters.

Harmonic $\bar{x}_H = \dfrac{n}{\sum 1/x_i}$ for rates whose numerator (work) is fixed.

Geometric $\bar{x}_G = (\prod x_i)^{1/n}$ for ratios that compound or normalized values. Arithmetic for times, sizes and anything whose sum means something.

The mean of a rate should track total work over total time. If every program does the same work F, then the right average rate is $\dfrac{nF}{\sum t_i}$, which is the harmonic mean of the rates.

Weighted forms, with $\sum w_i = 1$, $\bar{x}_A = \sum w_i x_i$, $\bar{x}_H = 1/\sum \frac{w_i}{x_i}$, $\bar{x}_G = \prod x_i^{w_i}$. Always $\bar{x}_H \le \bar{x}_G \le \bar{x}_A$, equal only when all values are equal.

Spread

Range $= x_{\max} - x_{\min}$. Sample variance $s^2 = \dfrac{\sum (x_i - \bar{x})^2}{n-1} = \dfrac{\sum x_i^2 - n\bar{x}^2}{n-1}$. Standard deviation s. Coefficient of variation COV $= s/\bar{x}$ (unitless). IQR $= Q_3 - Q_1$, SIQR $= \dfrac{Q_3 - Q_1}{2}$. Midhinge $= \dfrac{Q_1 + Q_3}{2}$. Use s with the mean, SIQR or IQR with the median.

Quantiles, course convention

Sort $x_{(1)} \le \cdots \le x_{(n)}$ and give $x_{(i)}$ the plotting position $f_i = \dfrac{i - 0.5}{n}$. For a quantile q, find i with $f_i \le q \le f_{i+1}$ and interpolate linearly between $x_{(i)}$ and $x_{(i+1)}$. Histogram bins by Sturges, $k = \lfloor 1 + \log_2 n \rfloor$ on the slide (HW3 rounded instead, so say which you used). Excel's PERCENTILE on the slides interpolates at $1 + (n-1)p$, which gives slightly different quartiles.

Normal Q-Q plot

Plot $(z_i,\ x_{(i)})$ with $z_i = \Phi^{-1}(f_i)$, or rescale to $m + s\,z_i$ so both axes share units. A straight line means normal. Ends bending away from the line mean heavy tails, and a curve bending one way means skew.

3. Confidence intervals (Lecture 6)

Mean

Large n (slides say $n > 30$) $\quad \bar{x} \mp z_{\alpha/2}\dfrac{s}{\sqrt{n}}$.

Small n, normal population $\quad \bar{x} \mp t_{[\alpha/2,\,n-1]}\dfrac{s}{\sqrt{n}}$.

All decks use upper-tail notation, $z_{\alpha/2}$ or $z[\alpha/2]$ is the value with area $\alpha/2$ above it, which the CI deck also calls the $(1-\alpha/2)$-quantile. Same number, e.g. 1.96 for 95%. The p.50 table labels them $z_{0.95}$, $z_{0.975}$. For $n = 15$ use df 14, $t = 2.145$.

Common levels $\quad 90\% \to 1.645$, $95\% \to 1.960$, $99\% \to 2.576$.

Batch means split a correlated series into batches and treat batch means as the sample. A 95% CI means 95% of intervals built this way cover μ. It does not mean μ has a 95% chance of being in this particular interval, and it says nothing about 95% of the data. The width shrinks like $1/\sqrt{n}$, so halving it takes four times as many runs.

How many measurements

For a half-width of e as a fraction of the mean, $n = \left(\dfrac{z_{\alpha/2}\,s}{e\,\bar{x}}\right)^2$, rounded up. Slide example $s = 2.14$, $\bar{x} = 7.94$, 90%, $e = 0.035$ gives $n = 160.5 \to 161$. The slide's "within 7%" is the full width, so the half-width is 3.5%.

Proportion

$p = m/n$, $\quad p \mp z_{\alpha/2}\sqrt{\dfrac{p(1-p)}{n}}$, valid when $np \ge 10$ (slide) and $n(1-p) \ge 10$ (study guide). Wald interval, poor for small n. Sample size with e as a RELATIVE error (slides) $n = \dfrac{z_{\alpha/2}^2\,p(1-p)}{(e\,p)^2}$. Slide example $e = 0.005$, $p = 0.1097$ gives 1.25 million 10 ms samples, 3.46 hours. With an absolute half-width e (measurement study guide) drop the p in the denominator. Slide example $m = 658$ of $n = 6000$, so $p = 0.1097$ and the 95% CI is 10.2% to 11.8%.

Variance (normal data only)

$\left(\dfrac{(n-1)s^2}{\chi^2_{1-\alpha/2,\,n-1}},\ \dfrac{(n-1)s^2}{\chi^2_{\alpha/2,\,n-1}}\right)$. The larger χ^2 quantile goes in the left denominator.

4. Hypothesis testing (Lecture 9)

Steps state H_0 and H_1, choose α and one or two sides BEFORE the data, compute the statistic, compare with the critical value or find p, then report the effect with an interval.

$Z = \dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$, $\quad t = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$ with $n-1$ df. Two-sided, reject if $|Z| > z[\alpha/2]$. One-sided, reject if $Z > z[\alpha]$. A two-sided test at α rejects exactly when the $(1-\alpha)$ CI excludes μ_0.

p is the probability of a statistic at least this extreme IF H_0 is true. It's not the probability that H_0 is true.

	H_0 true	H_0 false
Reject	Type I (α)	correct (power $1-\beta$)
Don't reject	correct	Type II (β)

Runs needed to detect a difference δ $\quad n = \left(\dfrac{(z[\alpha/2] + z[\beta])\,\sigma}{\delta}\right)^2$. At $\alpha = 0.05$ and power 0.80, $(1.96 + 0.84)^2 \approx 7.8$, so $n \approx 7.8\,\sigma^2/\delta^2$.

Not rejecting is not proof of no difference. With m tests at α each, expect about $m\alpha$ false positives, and Bonferroni uses α/m per test. Holm sorts the p values and compares the k-th smallest with $\alpha/(m-k+1)$, stopping at the first failure. Benjamini-Hochberg compares the k-th smallest with $k\alpha/m$ and controls the false discovery rate. Correlated samples count for less, e.g. 400 samples with $n_{\text{eff}} = 100$ halve Z. Statistical significance says nothing about whether the effect matters, so give the effect size and its CI.

5. Comparing two alternatives (Lecture 10)

Pick the test

1. Same unit (workload, benchmark) under both? Yes means paired, no means unpaired.
2. Paired with roughly normal differences, paired t on the differences.
3. Paired with outliers, Wilcoxon signed-rank if symmetric, sign test or bootstrap if skewed.
4. Unpaired and roughly normal, Welch's t.
5. Unpaired and skewed or ordinal, Mann-Whitney U or bootstrap.

For paired tests, normality is about the differences, not A or B.

Paired

$d_i = a_i - b_i, \quad \bar{d} \mp t[\alpha/2,\, n-1]\dfrac{s_d}{\sqrt{n}}, \quad n$ is the number of PAIRS.

Pairing removes the shared variation $2\rho\sigma_A\sigma_B$ from the variance of the difference, so it helps whenever $\rho > 0$.

Unpaired, Welch

$$s = \sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}}, \quad (\bar{x}_A - \bar{x}_B) \mp t[\alpha/2,\, \nu]\, s,$$

$$\nu = \frac{\left(\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}\right)^2}{\frac{(s_A^2/n_A)^2}{n_A - 1} + \frac{(s_B^2/n_B)^2}{n_B - 1}}$$

Use ν as is or round DOWN (more conservative). It lies between $\min(n_A, n_B) - 1$ and $n_A + n_B - 2$. Pooled t assumes equal variances and uses $n_A + n_B - 2$ df. Welch is the default.

Reading results

If the CI for the difference contains 0, you can't say they differ at that level. Overlapping single-system CIs do NOT mean no difference (HW6 Miller example), and checking overlap is not a test. Runs needed paired $n = \big((z[\alpha/2] + z[\beta])\, s_d/\delta\big)^2$, unpaired per group $n = 2\big((z[\alpha/2] + z[\beta])\, s/\delta\big)^2$.

6. Measurement (Lecture 11)

A CI describes sampling error and nothing else. Repetition shrinks random error but never systematic error such as bias, a wrong clock or perturbation. Strategies event counting gives exact totals with tiny overhead but no context, tracing gives full detail and order but the most perturbation and data, sampling sets the overhead you choose but is statistical and misses rare events. Exactness, context and low overhead, you get two. Report the clock and its resolution. Finer observation means more perturbation. Timer rollover after $(2^n - 1)\, T_c$ for an n-bit counter with tick T_c. A single reading is off by up to one tick, so time k repetitions and divide, $(T_k - T_{\text{loop}})/k$. Overhead share of a measurement is about $2c/(T + 2c)$ for read cost c.

7. Research practice

Plagiarism per Luke's TR includes copied text without quotes even when cited, close paraphrase, reused figures, and self-plagiarism of your own published text. Cite ideas, quote words. Authorship goes to people who made an intellectual contribution.

Student t critical values, upper-tail area

$t[\alpha/2,\, \nu]$ for a two-sided $(1-\alpha)$ CI, e.g. 95% uses the .025 column, 90% uses .05.

df	.10	.05	.025	.01	.005
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169
11	1.363	1.796	2.201	2.718	3.106
12	1.356	1.782	2.179	2.681	3.055
13	1.350	1.771	2.160	2.650	3.012
14	1.345	1.761	2.145	2.624	2.977
15	1.341	1.753	2.131	2.602	2.947
16	1.337	1.746	2.120	2.583	2.921
17	1.333	1.740	2.110	2.567	2.898
18	1.330	1.734	2.101	2.552	2.878
19	1.328	1.729	2.093	2.539	2.861
20	1.325	1.725	2.086	2.528	2.845
21	1.323	1.721	2.080	2.518	2.831
22	1.321	1.717	2.074	2.508	2.819
23	1.319	1.714	2.069	2.500	2.807
24	1.318	1.711	2.064	2.492	2.797
25	1.316	1.708	2.060	2.485	2.787
26	1.315	1.706	2.056	2.479	2.779
27	1.314	1.703	2.052	2.473	2.771
28	1.313	1.701	2.048	2.467	2.763
29	1.311	1.699	2.045	2.462	2.756
30	1.310	1.697	2.042	2.457	2.750
35	1.306	1.690	2.030	2.438	2.724
40	1.303	1.684	2.021	2.423	2.704
50	1.299	1.676	2.009	2.403	2.678
60	1.296	1.671	2.000	2.390	2.660
80	1.292	1.664	1.990	2.374	2.639
100	1.290	1.660	1.984	2.364	2.626
120	1.289	1.658	1.980	2.358	2.617
∞	1.282	1.645	1.960	2.326	2.576

Square roots

2	1.414	27	5.196	52	7.211	77	8.775
3	1.732	28	5.292	53	7.280	78	8.832
4	2.000	29	5.385	54	7.348	79	8.888
5	2.236	30	5.477	55	7.416	80	8.944
6	2.449	31	5.568	56	7.483	81	9.000
7	2.646	32	5.657	57	7.550	82	9.055
8	2.828	33	5.745	58	7.616	83	9.110
9	3.000	34	5.831	59	7.681	84	9.165
10	3.162	35	5.916	60	7.746	85	9.220
11	3.317	36	6.000	61	7.810	86	9.274
12	3.464	37	6.083	62	7.874	87	9.327
13	3.606	38	6.164	63	7.937	88	9.381
14	3.742	39	6.245	64	8.000	89	9.434
15	3.873	40	6.325	65	8.062	90	9.487
16	4.000	41	6.403	66	8.124	91	9.539
17	4.123	42	6.481	67	8.185	92	9.592
18	4.243	43	6.557	68	8.246	93	9.644
19	4.359	44	6.633	69	8.307	94	9.695
20	4.472	45	6.708	70	8.367	95	9.747
21	4.583	46	6.782	71	8.426	96	9.798
22	4.690	47	6.856	72	8.485	97	9.849
23	4.796	48	6.928	73	8.544	98	9.899
24	4.899	49	7.000	74	8.602	99	9.950
25	5.000	50	7.071	75	8.660	100	10.000
26	5.099	51	7.141	76	8.718		

Standard normal $\Phi(z) = P(Z \le z)$

$P(Z > z) = 1-\Phi(z)$ and $\Phi(-z) = 1-\Phi(z)$. Critical values $z[.05] = 1.645$, $z[.025] = 1.960$, $z[.01] = 2.326$, $z[.005] = 2.576$, $z[.20] = 0.842$.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

χ^2 quantiles, $P(\chi^2_\nu \le x)$

df	0.005	0.025	0.05	0.95	0.975	0.995
1	0.00	0.00	0.00	3.84	5.02	7.88
2	0.01	0.05	0.10	5.99	7.38	10.60
3	0.07	0.22	0.35	7.81	9.35	12.84
4	0.21	0.48	0.71	9.49	11.14	14.86
5	0.41	0.83	1.15	11.07	12.83	16.75
6	0.68	1.24	1.64	12.59	14.45	18.55
7	0.99	1.69	2.17	14.07	16.01	20.28
8	1.34	2.18	2.73	15.51	17.53	21.95
9	1.73	2.70	3.33	16.92	19.02	23.59
10	2.16	3.25	3.94	18.31	20.48	25.19
11	2.60	3.82	4.57	19.68	21.92	26.76
12	3.07	4.40	5.23	21.03	23.34	28.30
13	3.57	5.01	5.89	22.36	24.74	29.82
14	4.07	5.63	6.57	23.68	26.12	31.32
15	4.60	6.26	7.26	25.00	27.49	32.80
16	5.14	6.91	7.96	26.30	28.85	34.27
17	5.70	7.56	8.67	27.59	30.19	35.72
18	6.26	8.23	9.39	28.87	31.53	37.16
19	6.84	8.91	10.12	30.14	32.85	38.58
20	7.43	9.59	10.85	31.41	34.17	40.00
21	8.03	10.28	11.59	32.67	35.48	41.40
22	8.64	10.98	12.34	33.92	36.78	42.80
23	9.26	11.69	13.09	35.17	38.08	44.18
24	9.89	12.40	13.85	36.42	39.36	45.56
25	10.52	13.12	14.61	37.65	40.65	46.93
26	11.16	13.84	15.38	38.89	41.92	48.29
27	11.81	14.57	16.15	40.11	43.19	49.64
28	12.46	15.31	16.93	41.34	44.46	50.99
29	13.12	16.05	17.71	42.56	45.72	52.34
30	13.79	16.79	18.49	43.77	46.98	53.67
40	20.71	24.43	26.51	55.76	59.34	66.77
50	27.99	32.36	34.76	67.50	71.42	79.49
60	35.53	40.48	43.19	79.08	83.30	91.95